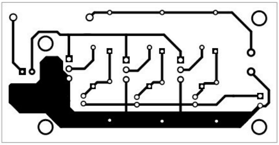

Fig. 4: Actual-size PCB layout of the Li-Fi dongle

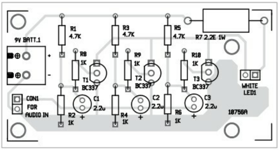

Fig. 5: Components layout for the PCB

connect the terminals of the solar panel to a 3.5mm female audio socket. Then, connect the 3.5mm audio jack from the speaker to the female socket.

During testing, Boom KBS M2 speaker was used. It is available on Flipkart at the following link: *https://www.flipkart.com/boom-kbs-m2-portable-mobile-tablet-speaker/p/itmesykcngmzayz6?pid=ACCESYKC8T6NUZTQ&otracker=pp_productCard_productRecommendation/browsingHistory_5*

However, you need not buy the exact speaker or Li-Fi speaker. Any speaker with inbuilt amplifier will do the job. You can modify the same to work as a Li-Fi speaker by interfacing a solar panel to it. After adjusting the band equaliser in the audio source and the distance between the solar panel and the light source, you can get almost noise-free sound from the speaker.

The maximum distance/range between the transmitter and receiver depends on the number of amplifier stages. For a three-stage CE amplifier, the maximum distance is about two metres with LED illuminance of 13,842.00 lux at the receiver side. For a five-stage CE amplifier, the distance can be up to 2.6 metres with LED illuminance of 17,382.00 lux. It was observed that as you increase the number of CE amplifier stages connected in parallel in the transmitter circuit, the intensity of LED light also increases, which, in turn, increases the maximum distance up to which the sound is audible from the Li-Fi speaker.

This circuit finds following applications:

1. Stream music from a mobile phone, MP3 player, laptop, tablet or other mobile audio system wirelessly to Li-Fi speakers. A camera flashlight can be used for the purpose.

2. Transmit audio signals from a microphone on the dais to speakers in an auditorium using pre-installed LED lights.

3. Use reading lights in buses, trains, cars and airplanes to play melodious music wirelessly on Li-Fi speakers.

The system can be used in offices, hotels, auditoria, etc, where bright lighting is required throughout the day.

Construction and testing

An actual-size PCB layout for the Li-Fi dongle is shown in Fig. 4 and its components layout in Fig. 5. After assembling the circuit on the PCB, enclose it in a suitable box and mount LED1 on front panel of the box. Connect the dongle to an audio source like mobile phone.

When you play the song, the white LED (LED1) will start blinking. Focus the LED light towards the solar panel. You should be able to hear the song from the speaker. Adjust the distance between the light source and the solar panel until you hear clear sound from the speaker. EFY

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Kit of this project is available at www.kitsNspares.com